

#121 - Unifying the Bayesian Brain Model and Free Energy Principle: A Dialogue with Karl Friston

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[Music] hello welcome to convergent dialogues this is xavier benia okay on this episode i am very pleased and honored to have the extremely brilliant carl fristen carl is a theoretical neuroscientist and pretty much one of the major authorities on brain imaging he's a professor at the institute of neurology at university college london consultant for the national hospital for neurology and neurosurgery in the united kingdom he has invented many aspects of brain imaging including the statistical parametric mapping voxel-based morphometry and dynamic causal modeling much of these were used in schizophrenia research which we talked about he talks about this in terms of his as his day job and then he has his kind of weekend job which is trying to find these theoretical models and structures of the human brain um and one of these theoretical neurobiological models is what he titles the free energy principle which we talk much about which also has some connections with the bayesian brain model he is a fellow of the royal society of biology and he has many honorary doctorates and we have a wide ranging conversation about many many many topics this conversation is over three hours um and i was absolutely you know privileged to to get three over three hours with with him he was very generous with his time and his his energy and his mind and we we talk about everything we first start by talking about the history of models of the brain so over time there have been many models of the brain and so he gives a history of what that looked like we talk about the evolutionary components of the brain and how executive functioning and abstraction work with various organisms we give an overview of the bayesian brain talk about predictive coding and or bayesian filtering we talk about the two classes of neurons for prediction error and for predictions he gives a very very nice comprehensive explanation of the free energy principle talk about the importance of a markov blanket and the importance of equilibrium and homeostasis we talk about the free energy principle and a little bit of freudian psychoanalysis and some of the connection there you talk about his views on consciousness and how active

inference is in play we talk about a.i we talk about the various neuroimaging techniques that i mentioned earlier we talk about fmris and some of their challenges there we also then end the towards the end of the conversation with how the free energy principle could be used pragmatically um i have to say this conversation is um it's long but it's it's very engaging i found the conversation to be very engaging and and even though it's it's pretty dense in some in some areas i've found it pretty easy to kind of track and follow along so you know carl does a really nice job of explaining very hard and difficult concepts in pretty easy to understand ways without losing the integrity of of the concepts and so it definitely requires some attention because these are some really complicated ideas but i think it's super important um he's doing really important work both in his day job uh and in the imaging for for the brain and then also his his weekend java as he says for trying to understand a theoretical framework and to understand that the neurophysiology and neurobiology of how things work together and how this can interact with engineering ai kind of computational neuroscience neuropsychology etc so again it was a an absolute uh honor and privilege to to have him on the podcast and he gave me so much time and energy and so i'm very pleased to bring you carl first him i'm here with carl fristen carl it is a big honor and a privilege sir so i thank you so much for coming on the podcast and uh being willing to talk to us about your very important work and all the things you've been researching so big thanks to you well it's a pleasure to be here thank you for asking me of course um so for folks that don't know who you are kind of give your potted biography just give us the snapshot of who you are um what you do uh what you study and all of the all the particulars right i have my day job and my day job is a professor of neuroscience at university college london um with a special expertise in human brain scanning so i originally trained as a psychiatrist um and got into the field of imaging neuroscience in an attempt to understand disorders like schizophrenia by literally looking at the brain as best we could or as best we can using modern technology and then i have a weekend job which is more of a theoretical neuroscientist and laterally one could generalize that to being a theoretical biologist or a biomathematician trying to understand the fundamentals of sentient behavior the principles that underlie the way that we make sense of the world that informs and is informed by the day job so the more you know about how what the brain must be doing uh the more you can start to look at the empirical results from the brain imaging and understand the message passing and the fluctuations and neural activity in the service of sensemaking yeah no this is very very important um i i guess the i do want to ask you about the imaging piece how did you kind of i guess um i guess stumble into this kind of way of doing the day job and the weekend

job this is something that you mentioned you were trained as a psychiatrist but was this something that you always wanted to do you were always curious about the brain and looking at different ways of understanding its functioning and then did the theoretical stuff kind of come later or how did this kind of i guess morph into where you're at now uh and what you're currently doing that's an excellent question which i could answer quickly or take a very long time answering the question whichever way you want let's do the quick one i always wanted to do this uh so what i am doing now is really the denouement of a series of uh serendipitous of lucky career choices that actually started around the age of 15 wanting to be what in those days would have been known as a mathematical psychologist so a dual commitment to understanding the most interesting thing which of course was me and my brain at that time and other people's brain but using a formal calculus uh your principle approach so trying to maintain a training and investment in physics and mathematics a sort of dual early career in psychology and physics and then went into psychiatry as the most alluring and i have to confess the easiest way to become a medic at that point in time because everybody else wanted to be a heart surgeon at that time it happens everybody wants to be a gp nowadays but my cohort they all wanted to be heart surgeons um so but i ended up in psychiatry and and very much valued that experience but of course it was exactly in the direction of travel um in terms of what i ultimately wanted to do which was research into what would nowadays i guess be thought of as um theoretical neurobiology or computational neuroscience but essentially the formal principles that underwrite our ability to engage with the world and make sense of that world yeah it's interesting the way you put it because i think that finding out kind of how you as you say there's this serendipitous way of falling into these things uh which is i think kind of true for a lot of people right they they have a one way in which they want to do things when they're younger and then they take the long way up the mountain but the the other thing is is having a desire passion to want to do something to understand the brain and many people want to understand the brain but then having the iq points and the capacity to do that and you've written some pretty prolific papers and so you know having the ability to to do the equations to do the math to be able to do these things that had to all of just when it kind of fit together you know had to have been pretty remarkable i guess for yourself to be like you know what i would like i have this interest and i can actually cognitively do this i can figure this out what has that i guess just been like as you you know progress you know through your life and be like you know what yeah i've i've done a few things i've i've researched a few things i've figured some things out about the brain if you kind of look back on that how does that just all

kind of sit with you i guess i would imagine it's the same kind of um sense that people have when they look back on your on their lives it's a long journey and when you look at the journey in retrospect you are sometimes surprised by how far you have come because in pursuing at any one moment in that journey it all seems very incremental so it seems unremarkable during during the travel unremarkable in the sense that things change quite slowly and are the reflection of the way that you've invested in you know the early parts of the of the journey so coming back again to this um early commitment to um a formal way of expressing ideas so you know knowing that meant that you had to spend many years um learning quantum physics and probability theory which then you immediately forgot as soon as you went into psychiatry at least at least you left with the legacy but you knew you could learn it and you couldn't do it if necessary and of course uh in later life it didn't become uh necessary um but i repeat i've forgotten everything i learned as a 20 year old but with the uh the boon the gift of wikipedia i was soon able to get back up to speed again so all of this prolific writing um that uh is often claimed to be quite inscrutable and mathematically uh difficult to access uh from my perspective is a nonsense you can get all the maths you need uh from wikipedia because that's exactly where i go it's the power of technology is remarkable in many ways so that's great so obviously i want to get to your your main uh your main i guess theoretical claim to fame if you will i mean many claims of fame but i definitely want to talk about the free energy principle and just have you download that for us and just tell us everything about it um in as much detail as you'd like i figure before we we get to that it might be helpful and you can tell me what you think of this it might be helpful to kind of look at a history and overview of various brain models that we've had and how leading up to the the current brain model that we know and then we can branch off to that into the kind of bayesian brain model which which is also uh more relevant and salient at the moment so maybe if you could just in your in your from your viewpoint how do we understand a history of various brain models where does it kind of first start and originate and you know you can just kind of kind of march us through how that gets us to current day and what the current brain models are well yeah that in fact that would be i think um one fairly comprehensive and complete way of understanding the free energy principle which is just a genuine of a long legacy of ideas that i think you could probably trace back to the students of plato and shadows on caves and the like but um for me most clearly articulated by helmholtz in the 19th century and sort of polymath who had a deep understanding of physics but also perception and you know various aspects of of sentience and many of his ideas essentially summarized in the form of unconscious

inference so the notion that the brain was a constructive organ that was inferring the causes of its sensations so the idea here is that what is out there is not directly accessible from the brain the skull-bound brain but has to be hypothesized you have to have the cause of your sensations in mind in order to use the sensory impressions upon your eyes your ears and your skin as evidence for against that hypothesis so as andy clark would put it this is very much an inside out approach to perception as opposed to an outside in view which actually dominated in the 20th century but then went away again where the the world impresses itself upon your sensory epithelia and you somehow extract information from that to get to the conclusion oh this is the cause of my sensations helmholtz had a much more nuanced and vertical or certainly by 21st century standards vertical understanding of the mechanics of perception you had to actually generate the hypothesis and you just used the sensory evidence at hand to update your beliefs adjust your explanation for the sensations and in that adjustment in the updating on the inside through neural activity you were able to produce a an approximate or partially faithful reproduction of what could have caused your sensations and he called that unconscious inference that notion remained in advance through 20th century behaviorism manifesting in many different ways things that come to mind include analysis by synthesis one key contribution from psychology was the notion of perception explicitly as hypothesis testing this is richard gregory a uk psychologist's notion again reiterating the importance of inference so for helmholtz it's really unconscious imprints um for gregory it would have been you know the perceptual act is basically a test of a hypothesis that this visual object is the best explanation for this pattern of sensory impressions say on my on my retina those ideas um came to fruition in um machine learning and computational neuroscience i would imagine the 80s and possibly the 90s probably best um exemplified by uh the helmholtz machine deferring back to helmholtz's original ideas uh by uh geoffrey hinton and colleagues and peter diane and colleagues so leveraging those deep insights from helmholtz and then putting them into a recurrent neural network or a network of a particular kind of kind of boltzmann machine effectively training neural networks in a way that we would now understand the training of deep neural networks in deep learning for example to infer we now understand that as amortization if you were in machine learning it would be this notion that you can train the connectivity of a neural network to make sense by inferring the causes of its inputs because you know it's a simple move to understand this schema algorithm of the helmholtz machine as also applicable to neuronal networks and at that point in time people were converging on very similar um from a slightly different direction they made that a predictive coding so predictive

coding was originally devised in engineering as the most efficient way to compress sound files in the 1950s and there's an interesting connection here between the notion of compressibility and efficiency and the statistical efficiency of a good inference which you may well come back to in a moment. I mention the conversation but at this stage in terms of the history and the legacy, there was in the 90s then a key convergence on the brain as an inference machine and the brain as a predictive machine so we then see the emergence of predictive processing or predictive coding in particular perhaps just for your listeners just to make this a bit more explicit the predictive coding is defined in terms of a scheme that is trying to resolve prediction errors so we have this notion now of the brain generating top-down predictions on the basis of its currently preferred beliefs about the state of the world those predictions then compared with the censoring input and then the mismatch or the divergence between the predictions and the observations the sensory samples are then denoted as a prediction error and then that prediction error is a broadcast deep into the brain's hierarchy to update and revise the hypotheses or the representations that are generating the predictions and I use the word generation there because all of this machinery all this message passing with sense making comes under a particular generative model that is entailed by the architecture of the brain so you have this picture now of the brain broadcasting from the inside out descending down the hierarchy it's predictions and there's a counter stream of ascending prediction errors that are revising and updating the uh the predictions in virtue of changing the representations that can also be cast in bayesian terms as bayesian belief updating so now you've got another church of mathematicians theoreticians in the game you've got you've got the helmholtz machine you've got predictive coding people from engineering and also people start people like simon mclaughlin and raj rayo and darla ballard um noted that these predictive coding principles were very apt to explain a lot of the functional architectures of the brain david mumford i think was a key contributor to those ideas in the early 90s and it's not usually acknowledged as such but you know if you go back in very early 90s uh you know very explicit and what proved to be a visionary account of the neuronal architectures that did the job of the health health machine that did predictive coding did hierarchical bayesian influence so then you have the joint emergence of helmholtz machines machine learning the bayesian brain hypothesis um the neurosciences and predictive coding as a particular process theory that sort of enacted this so that was a story at the end of the 20th century um at which point there was an enacted term a turn back to the four e's the um the importance of situated cognition the importance of the fact you have a body that your perception is an active thing

you act you actively palpate the world at every level every sensory modality whether it's literally feeling a way around a dark room to confirm your hypothesis about this is a telephone this is the bed this is my iphone through to palpating the world visually through active vision moving your eyes around something this part of the visual scene to test this hypothesis of that hypothesis so this um embodied inactivist extended um and by extended hair i uh you know one's acknowledging that a lot of your cognitive abilities are um downloadable into the environment the key thing here though is this uh inactive term so how did predictive coding the bayesian brain hypothesis um survive the inactive turn well very very simply that there are two ways that you could effectively resolve your prediction errors you could either literally change your mind in the good old-fashioned predictive coding where you use your prediction errors to drive a change of mind amazing belief updating so the predictions now were more fit for purpose and explaining where the sensory inputs and thereby the prediction as effectively canceling themselves resolving themselves through this hierarchical message passing in the brain or and here's the twist you could literally go and re-sample the sensory input to make it match your top-down predictions but now you have a really simple and a very plausible account of active sensing active vision uh more generally active inference the you know there are two ways in which you can resolve prediction errors you can either change your mind or change the thing you're predicting and thereby supplying the predicted outcomes so this is a you know a very simple account of action and perception and also um speaks to the circular causality you know these are two processes working hand in hand your perception is causing action and action is causing perception and this is how we sense make so that um that sort of active inference perspective um which then i think um became sort of the dominant paradigm the cognitive neurosciences under the rubric is something called predictive processing which was a sort of generalization of predictive uh predictive coding to accommodate this inactive aspect uh of perception so the free energy principle or on one reading really is just a mathematical specification of this kind of active inference um the trick um here is um to cast things in a most general and generic way as possible and by things i just mean the resolution of prediction errors um and the way that one does that is by appealing to the maths of bayesian inference and noting that exact bayesian inference is not physically realizable and you then have to turn to how can you physically realize inference of this sort and the solution was in fact um uh on offer from uh you know in the 1950s from people like richard feynman sort of western um physical sciences um but also hints at exactly the same solution um in the early russian literature in terms of algorithmic complexity and comorograph complexity those two

perspectives converge upon exactly the same mathematical quantity which is a variational free energy this is not a thermodynamic free energy this is just a statistical way of measuring thing that you want to optimize in any act of inference whether it's your brain doing inference about its sensations or whether like richard feynman you're trying to infer the probability distributions of the paths of small particles in quantum electrodynamics uh whether you're trying to infer the message that was sent in cryptography for example all of these kinds of problems could be subsumed under one optimization process which is the maximization of bayesian model evidence it's literally likelihood of these data being generated by the model that you're currently considering so that's in basic systems known as bayesian model evidence and the the variational free energy provides a tractable bound or approximation to that marginal likelihood or moral evidence which means that in its most general form the problem of and indeed the process of inference you know if you don't want to consider it a problem can be written down mathematically as any process that either maximizes marginal likelihood model evidence or if you're a machine learning maximizes free energy as an evidence lower bound also known as an elbow elbow there's a slight wrinkle here in machine learning um they use the negative of free energy that feynman used so if you ever talk to physicists that you would want to minimize your free energy from talking to machine learning people you want to maximize it it doesn't matter from our point of view the key thing is that you're just writing down the functional the mathematical form of the quantity that underwrites the process of inference that is a variation of free energy in optimizing either minimizing or maximizing this either by perception or through action then you are effectively self-evidencing and that's a term which i like from jakob that comes from philosophy that if we understand ourselves as making sense of through a process of inference that implicitly what we are saying is that we are all in the game of garnering sensory evidence for our models of that world so we are literally self-evidencing and you can just write that down as the process of maximizing bayesian model evidence for your model entailed by your brain and your body or you can write down as a physicist in terms of minimizing a variation free energy and that's the free energy principle and that's roughly where we stand at the moment there are lots of unanswered questions about construction you laid out uh very very nicely i was it's very easy to follow there i have i do have many follow-ups here um about a few things i'll try and keep them very general and kind of reframe some of what you said so starting with hemholtz right you were you were talking about how there's this inside out outside in right that there's almost a kind of dichotomy in how we understand the world right do we understand it from within ourselves

and then based on what you know input outputs outputs input whatever way you want to describe it is it and maybe that's just how it's falling out is it always kind of in these two uh camps or categories or is there a third or fourth way is there another way where we could see a kind of mixture where it's not exclusively outputs from the environment excuse me inputs from the environment and then we output is there some way in which we can understand that there's a third way or a fourth way or other ways in which these are combined and where it isn't just input output right it's not just a kind of zeros and ones kind of thing is there another way in which there's uh an integration of them or or potentially some way of the phenomenology of how we're experiencing things is it is it always this kind of way and how it lays out because it seems throughout time as you were saying with hemholtz there's this um there is this kind of dichotomy where people will kind of look at the the economy here and then all the way to the machine learning stuff in the 90s it kind of does this oscillating back and forth between where do we put the emphasis where do we put it first so i guess that's one that's one component uh the second uh piece here when you were talking about if i understand this the free energy principle from what you're saying sounds like it's the kind of quantitative and statistical ways of understanding the bayesian model of the brain and so they're kind of seen in tandem along with the active inference which you which you explained connected with that there's this emphasis on perception especially with humans so i wonder with the predictive types of elements that you were discussing where do we this might be a little bit further out and maybe some of this may be hard to know all of this is very human i wonder if we understand some of these things for non-human brains as well such as with chimps or bonobos or or even other types of it might not even be with primates it could be what do we understand about how potentially these workings of the brain could be for other brains that are non-human so trying to look at a kind of evolutionary structure and model and then finally um uh curious about if you have any thoughts about you mentioned philosophy and as you were saying that there was this kind of in the 21st century now not just an over emphasis on the brain but how the brain and how it works is distributed through the body and how the body is a necessary component which automatically always makes me think of uh philosophy of merlou ponti who also thought a lot about perception and in many ways had very technical ways of thinking about perception he talked a lot about eyes he talked a lot about sensations and i wonder his estimation he was looking at the phenomenology of things of how do we understand experiences through the body that the body is a necessary or or is a prerequisite of sorts as a type of bridge for understanding our various perceptions um so those are my follow-

ups i don't want to throw it all at you at once but any any thoughts about any of that i guess right you you've just thrown all three massive issues at once yes yes those wonderful we have to say at least something about each one of them yes please so in order then so the um is there a third way that does not um commit to an inside out um information extraction or sorry an inside out construction versus a constructive constructivist approach versus an outside in extractionist approach and yeah in a sense that just is the bayesian brain that just is um the free energy principle and active inference that is entailed by the free energy principle because that is explicitly saying you have to have this two-way traffic and this emerges in you know many different levels so for example um we've already talked about this sort of recurrent um two-way traffic of message passing in the brain from the sensory parts of the brain to the deeper hierarchically organized um levels of a the brain's generative model this whole belief updating this whole bayesian mechanics depends upon this counter stream of top-down predictions and bottom-up bottom-up prediction errors on a predictive coding architecture or view um so you've got this recurrence you've got the circular causality for free but even more generally as as you intimated in the third issue um you brought to the table here the the you know the very um dependency of perception on action and action on on perception so put simply um i have to perceive and infer in order to plan what to do next so action is predicated on my perceptual synthesis and making sense of the current state of my body and the world and what i want to query the world about the you know what kinds of questions what kind of information do i want to get by deploying my uh sense organs that is you're almost self-evident but in the same way that actually depends upon perception perception also depends on action what i see depends upon where i look or you know how i place my um skin in relation to an object that i'm trying to palpate so there's this unbreakable um circular causality and two-way street between action and perception so i would submit that your your quest for the third way as has has been delivered for you by um by the inactivist term by the an activist application of predicted uh your two predictive coding to the brazilian brain hypothesis um at the turn of the century i think you know that is the third way you can't tease these two things apart really um so that that should be comforting um and it should resolve any angst that wouldn't be seen it's very satisfying yes now the second issue what was the second issue the second one was about about the in non-human brains oh yes evolutionary construct yeah the way that you ask that question makes me realize i've made a fundamental mistake here in uh in terms of telling the the story and unpacking the narrative of um of the of active inference um in its 21st century version and so most of what i've just talked about in terms of um the mechanics of inference

and um optimal action could be applied to a thermostat or a virus i think i would actually put the question the other way around and perhaps that's how you actually meant uh the question i'm sure that we'll come to this later on um you know what is what is special about the human brain that um uh has some unique and definitive properties that you were not found in a thermostat that you would not find in a virus you would not find in a mouse you possibly wouldn't even find it in um in your most loved pet or you know some of our closer um uh companions and uh say monkeys you know an evolutionary um view um so i i think that that's a you know that's the the direction the question um should be approached it's less you know what's special about uh mice it's more how do you make the move from mice to men under this kind of architecture so most of the work that that we do is at a much lower level it's just trying to understand the fundamentals of active self-organization under through the lens of sense-making and imprints that can be written down as a process that usually is actually emergent uh an emergent property of any self-organizing system that preserves a distinction between itself and the environment that doesn't decay dissolve dissipate die um so you know looking really trying to explain how on earth does a virus survive what must it do in terms of this circular third way exchange between the inside and the outside to preserve its very boundaries and de-marketing from its external milieu without worrying too much about what it is to be human so we haven't got anywhere near as marx sounds like a set of questions yet we are not talking about um bayesian beliefs as propositional beliefs that you and i might hold and be able to articulate we are not talking about qualitative experience we are just talking about the mechanics of brief updating on neuronal circuits neural circuits chemical circuits electrochemical circuits that we see in every kind of biotic and possibly even non-biotic self-organization so that's a deep question um and the answers to that question take you through a series of natural kinds of things all are doing some um inference of a a possibly a weak kind um and can all be if you like um described in terms of an ontology of generative models if you remember before we're talking about the underlying importance of a generative model in supplying the architecture under which you generate your hypothesis your top-down predictions and you use basic this is the thing that you are securing evidence for which means that um if there is a difference between mice and men then there has to be something quintessentially different between the genitive models that you and i employ to navigate actively our world and those that might and flies um use um and once one starts to think about it you know one might ask what is the difference between a um a let's take viruses flies are a little bit too sophisticated what's the difference between a virus and a mouse and what's the

difference between a mouse and me or you um and normally the story goes that it's all to do with the ability to plan it may sound very very simple but you know once you've heard the argument you know it is actually a very simple argument but it's also quite a compelling um argument and has a degree of expansion scope in the sense that um the thing that distinguishes between a thermostat or a virus or some single celled organism succumbing to nutrition gradients through chemotaxis relative to something like a mouse that does stuff in anticipation of consequences is the ability of the mouse to plan the mouse can think about and select those actions that have consequences and that's absolutely profound when it comes to writing down or thinking about the genitive model that that mouse is using it tells you immediately that the mouse has a model of the consequences of its if it has a model of the consequences of its action given that the consequences occur after the action then it also has a model of the future so it now has a model that has a temporal depth and this can be contrasted with the thermostat or virus it does not the thermostat doesn't have a model of the future it doesn't have an explicit um hypothesis or question in its internal machinery of the kind well what would happen if i switch this heat making device on or not it just responds reflexively in the moment but the mouse will have a much more elaborate model sometimes described as a model with temporal depth that frees you from the moment but also crucially underwrites the ability to plan and thereby select from one plan or another plan one course of action relative to another course of action that endows that little agent with agency because now it is selecting amongst a number of different courses of that are entertained and can be supported by its genetic model so now together you've got this sort of um move from a thermostat to a mouse that entails the notion of agency and temporal depth simply because this generative moral has the ability to model the consequences of action so that's one move does that give you qualitative experience does that give you propositional beliefs does that do we have mouse philosophers who spend their time puzzling about why they see red and does one mouse see the same red as another mouse no absolutely not so we still haven't answered the question um what's the difference between a mouse and a man just sorry just on the on the mouse thing it sounds like what you're describing in the ability to have some uh planning sounds like there's at least a low level type of what we would call executive functioning though where there's abilities of plan organized sequence right which we normally see in the prefrontal cortex for humans but it seems like it's a very low level uh ability whereas as we get to humans there's a much less deeper level of abstraction that we understand is this is this about right now that's that's exactly and absolutely right so all i am talking about here um is exactly the the prefrontal

cortex um and all the attendant uh executive machinery uh right down through uh supplementary motor area premonitor courtesy metacortis down to the brain stem and all uh and all the um the executive components that are responsible for executing the plans that are elaborated through selecting the right course of action in the prefrontal cortex so that's exactly right and that's what mice can do um you know they don't have quite as well developed a prefrontal cortex as we do but they have the equivalent um you know the equivalent capacity to execute plans simply because they need to have that because they have plans in their head but having plans in their head is a remarkable capacity you know a thermostat i repeat does not need to have a plan uh you know it just responds as a reflex as do many you know forms of lower life or biotic self-organization so you also used um you know a hierarchically deeper levels of abstraction i think that's the absolute key here and and then this question would be well what are you hierarchically abstracting and i think most people would answer well you are now creating models of your executive process so now a mouse has the ability to plan and to execute those plans by selecting the plans that are most plausible in the sense that they will resolve surprise or prediction error or free energy in the future we call that expected free energy uh technically um which has some interesting aspects to it the the mouse then has this ability so what ability do we have because of our more evolved prefrontal cortexes with these deeper levels of abstraction well it could simply be to entertain the hypothesis that i am a person who is now selecting these plans and it is me who is actually executing these plans to a generative model that now entertains the hypothesis the belief the representation that it's me doing these things it is me who is embodied it is me who is the agent so an agent doesn't need to know that it is an agent to actually plan and execute those plans in the same way the thermostat doesn't know it's a thermostat and then it's even acting so i think that the levels of abstraction you're referring to here are exactly the abstraction and the plausible hypothesis that in fact i am an object and indeed i am the subject of my own observations and actions and having that hypothesis as part of the genetic model now has to be explained and how can it be explained how could it be that i now have a better generative model or explanation of my sensorium that has been actively constructed um simply by having this notion that i am a person and i think there are you know quite simple answers to that because of course it's not just i am a person i'm a person in this police state or that police state or that emotional state or this uh intentional proposition state and once i recognize myself to be in that state and implicitly now i have a degree qualitative experience of my inference lowering the hierarchy then i can now bring to bear and contextualize all my message passing and inference

conditional upon my now recognized state so for example if i know that i am anxious or i am embarrassed i know that there will be certain kinds of information to which i would be paying attention um there will be certain sort of autonomic actions that i would uh normally be engaging in this context of being embarrassed or being frightened such as cardio acceleration or paying particular attention to um auditory and visual streams for example so there is there is an easy to argue benefit of for having these higher levels of deep more deeply abstracted representations and explanations the active inference and all the message passing that's going on that's going on underneath and i think if you go right to the top of the hierarchy to the very highest form of human life which would be a philosopher what you find there are representations and hypotheses that just are qualia and qualitative experience so you know this is touching on now um something which david charmans has popularized as a the meta problem or the meta hard problem is why do we puzzle about our qualitative experiences you know ice don't do that so once special aspects of our genitive models can explain the very fact that we can entertain notions like being a zombie for example or thought experiments like brain in a vat the very fact that we can entertain these counter factual outlandish hypotheses about self is the gift if you like of having these deeply abstracted representations of how we engage with the world you know first of all just um having a representation of me or selfhood and then you know me as an actively inferring agent that must in some sense have some internal action over the way in which i secure the evidence and test my hypotheses i'm trying to work here to um assumption which i think mark uh sounds may have um talked about which is you know what underwrites the the feeling part the um the you know the quality of experience the philosophers talk about in terms of qualia um the um the the essence that we think we can uh um entertain and have discourse about that perhaps um a mouse might not be able to talk about it may have but he wouldn't be able to recognize it or talk about it it's not just not there it just can't see that kind of thing in the same sense that a blind person can't see color it may well be that a mouse just not does not have these hierarchical levels of abstraction so it's just not perceivable in the sense of helmholtz um you know you have to have it in mind before you can before you can you can sense it the argument the answer that somebody like mark sobres i think a lot of people would would bring to the table here is the notion of um confidence or precision or a representation of uncertainty in one's beliefs and getting that right and that basically entails a kind of mental action so now we're talking about action not on the outside not autonomic actions like secreting something or cardio acceleration and we're not talking about motor actions motor reflexes moving my limbs around with my stride and

muscles we're talking about a covert internalized whereas very very high levels of a hierarchy or very deep levels perhaps you know at the center of the brain from uh mark services perspective have an influence on the belief updating that underwrites the action selection and that influence is really statement that we're able to represent our own beliefs we have beliefs about beliefs the particular example here is um we have certain beliefs about our beliefs that pertain to the uncertainty so um if you have in mind the simple mathematical description of a belief as a probability distribution so when people talk about basically i repeat they don't they're not talking about sort of ideological or theological commitment you know they're just talking about um the the brain as a sentient artifact that is somehow encoding a probability distribution and invariably these kinds of beliefs these bayesian beliefs of conditional probability distributions have two cardinal aspects they have a location so that the most likely thing i believe or the basic belief entails designated by the peak of the probability distribution and there's another shape parameter um another aspect of this probability density or distribution which is its spread or dispersion and that's usually understood in terms of uncertainty so very broad distribution means that all of these things could be in play as a very sharp very precise distribution means that i'm pretty confident that it's just underneath this very sharp distribution so that precise that that precision is a really important attribute of a belief and if parts of the brain now have beliefs about beliefs they are now predicting precision predicting predictability and that may well be the thing that um calls for cover mental action and the need for recognizing that i'm in this belief state or this state of uncertainty on that state of uncertainty and then i can now deploy the uh the appropriate beliefs about beliefs or predictions of predictability or predictions of precision yeah it's interesting on that last point because what you're sounding like you're describing is this this type of metacognition that's interested in our own epistemology and also our ontology as well right that we're very concerned about our beliefs and how we how we have these beliefs about them or even things about our own existence or things about us who we are as beings one other point i wanted to make about the abstraction piece is that i totally agree obviously but i think even even more uh miraculous if you will or more wonderful is the fact that for humans not only can we have these levels of abstraction but i always think of this example of like you know i can know that i'm thinking about a concept or i can know that someone else is potentially thinking about a concept right there's this way of abstracting with the other but even further if you take works of fiction let's say let's say i'm reading a novel and the central character is thinking about a thought he had in a dream that he had there's and i'm the one reading that thought that

someone wrote you have like six layers of abstraction there right this is very big big big wide lens of things which is incredible it's incredible our mind doesn't you know fold up you know because of how complicated or complex that is it's in and i don't know if other animals on the planet can you know read novels and get the whole contract or if people watch a movie right and they can see or observe all of these things um so it's very interesting about the abstraction and i liked how you linked with the various components of executive functioning for for detailing that we've talked a lot about the bayesian brain and we've mentioned many components of it maybe we can give for for listeners the kind of overview of the bayesian brain and then that will kind of dovetail into the free energy principle and i'll let you just i'll give you a lot of runway for that you can you can talk all about that so my under you correct me here if i'm wrong my limited understanding of the bayesian brain is that the brain is essentially equipped with an internal model of the environment and that gives us a plan for how do we create sensory observer observations from hidden states right that's one way of saying it and that there are these hidden variables that are from a prior distribution you could say maybe i don't know if you want to use the freudian term the unconscious we can come to freud at some point um and then also that these sensory observations are have a distribution conditional on the hidden state so then you have these priors and you have a likelihood combined to infer the hidden state given the observations so i think the clarifying point here is that these aren't specific claims about priors and likelihoods for an individual but more so on the consistency of them that's my very limited scope of the bayesian brain so maybe kind of branch off of that and just kind of give us the full overview of what we mean by the basin brain priors likelihood hidden states and then we can jump into the uh the free energy right and well i mean that description was spot on in itself so all i need to do is to connect what you've just described okay what we're talking about i think that we basically have done the bayesian brain uh and we've done the finest and most delicate aspects of of the bayesian brain under a hierarchal genetic model when you started talking about the you know the six degrees of hierarchical embedding and i'm glad you used the word metacognition so that's what i was trying to imply i was talking about beliefs about beliefs this is exactly it and it is this hierarchical uh recursive embedding which is quite remarkable you know uh i can't really use the word fall down or unfold but but certainly you know at about six degrees of um of recursion uh my brain gives up and i think i'm at that point yes that may be the explanation for the better hard problem that you know even philosophers can't quite handle that and that residual puzzlement you know uh inspires us to try and get to the seventh or the or the eighth

but it is exactly that sort of bleed about beliefs about beliefs the other key thing you just brought to the table before um just unpacking the bayesian brain um the language and the setting that we've already established um you brought another really key thing to the table which is inference about others and i was tempted to say well perhaps that's one aspect of the third way that we haven't talked about that we haven't spoken about and perhaps we should have spoken a lot because everything that we've spoken about thus far would work for a single agent in a universe that did not have anything else like it in that universe it's just making sense of the universe um but of course most of the interesting questions um in terms of self-understanding are in light of or contextualized by the need to understand others so when you're we're now moving from the bayesian brain and beyond that to the embodied bayesian brain which is the third issue that we've still yet to talk about um but now to multiple bayesian brains in an ensemble all ex all trying to infer each other and that brings some really interesting questions to the table about self-understanding in relation to others understanding and there could be an argument the the need for selfhood or a an explicit ability to recognize i am a self and to have at least a minimal selfhood is just a product of being a being required disambiguate the causes of my sensations in terms of did i cause that or did you cause that or perhaps more practically um more relevant would be uh during very early early neonatal experience did did i cause that or did mother cause that so mother being the key other and this being able to first of all recognize mother is actually an object out there that is distinct from me is a great move to be able to do and something that takes you on sort of a long time for it for a neonate to and not every dna you know recognizes that and can learn the appropriate genetic models that explains interactions or our interactions with mum in terms of mum being a creature because once you realize that mum is a creature or an object out there then the natural hypothesis might be to enter well perhaps i am a creature and why would you need to do that well you need to do that between disambiguate did i cause that or did mom cause that so that elemental theory of mind the ability to distinguish between the causes between the causes of sensations in terms of self versus other may be unique to creatures that have a universe or an environment that is populated by creatures like them such as you and me so i think this is a really key um aspect of this sort of self-evidencing self-evidencing yes but it's in the context of a whole world comprised of things which are just like me so i have to now be very skilled at differentiating me from you and in so doing of course pressure is only convinced if you are sufficiently similar to me and cause the same kinds of sounds and relative touches and sensations that i could cause i don't need to do that given that you are very similar to me but if

you're that similar to me i can now use my genitive models my explanations of the world to explain uh which i would normally use to explain my autonomic dispositions and you know my bodily sensations all my movements i can now use those prior beliefs selections of the plans that we were talking about our intentions to explain your intentions or more specifically to explain your behavior your facial configuration your micro expressions your physical stance um in terms of your intentions which would be the intentions i would have to have in order to witness and you know to manifest the same kind of observable behaviors so i think that you know that's a really important thing you brought to the table and you know in terms of self selfhood and self-evidencing or being particularly acute um and important um in the context of your a world that has has a culture uh that is born of the fact that we are we populate that world with lots of con specifics and we have a we have a shared economy in which to do um i just wanted to put that on the table because it's a really important thing when it comes to sort of personal exchanges yeah absolutely but the back to the basic brain well you just summarized it i mean i don't you've done all the heavy lifting there i think it's probably best just to say out loud at this point there is nothing that the bayesian brain brings to the table we haven't already spoken about it's just a way one way of articulating um the kind of sense making where we have say prediction errors ascending the hierarchy from from the sensory cortex and sensory parts of our brain and updating that would be in a bayesian sense regarded as posterior beliefs so what i'm going to do now is just tell the bayesian brain story from the point of view as somebody who likes to talk about predictive coding so we've already established predictive coding predicted coding i should just add if i was talking to an engineer i wouldn't use the word predictive coding i would i would use the phrase bayesian filtering the simplest instance of a bayesian filter in something called a kalman filter which is used widely in state estimation predictive coding just is kalman filtering of an extended and non-linear form sometimes known as an extended kalman bucy filter of a hierarchical sort um so these things are all the same but different people like to use different language so what would how would you understand bayesian inference from the point of view of predictive coding well as you say bayesian inference is just the process of belief updating how would you articulate that from the bayesian brain hypothesis well that update is just an update to your bayesian beliefs your conditional probability distributions prior to seeing some data to some beliefs after or apostle so you're just updating your prior belief to your posterior belief you're literally changing your mind you're just moving through changing neural activity or weights or connection strengths in your brain uh you're changing implicit or encoded bayesian belief

from a prior belief before you saw the data to a posterior belief after you saw the and that movement from one belief to another belief literally changing your mind um is the updating that is driven by the prediction errors so the next question then would be well what is the nature of that updating um as we if you like mix together the prior belief and the likelihood information afforded by the likelihood of these data to produce the posterior belief so that's just to explain to the listeners why there were three kinds of beliefs there's all likelihood distribution um a prior belief and a posterior belief and really the likelihood um if you mix together the prior belief and the likelihood that you get the posterior out so the likelihood stands in for the um the pressure or the forces applied to changing your mind that drive a change in mind or the belief updating that are afforded by the data and this is i think a useful view um it allows us to come back to this notion of precision it's difficult to do uh without a white board or a black ball but let's have a go because this is i think a really intuitive and really important observation which brings us back there's a mental action of the notion of predictions of precision so if i had a very very vague imprecise prior belief uh and i got i had some very definitive very precise sensory evidence that could be of an elemental sort like particular pattern of visual impressions in a well-lit room or it could indeed a more abstract level be some definitive information from wikipedia but if it's very precise then that precise likelihood will have a very sharp narrow distribution and will pull the soft flat prior distribution much closer to the likelihood so that my posterior now is now dominated by the shape and the form of the likelihood so i've induced a much greater degree of belief updating i've changed my mind considerably more than i would have done simply because i ascribed a greater precision to these data so it had a much more precise likelihood conversely if i have a very very precise prior belief that it's based on lots and lots of experience then it's going to be much less sensitive to any likelihood or any my posterior is going to be if you like tethered to my prior belief as i change my mind and i move from my prior to my posterior area when seeing the evidence doing my basic belief update according to bayes rule i will move a much smaller distance from the uh from the prior belief so the precision afforded both the prior belief and the likelihood in fact it's the balance the ratio of the two becomes a really important quantity in any bayesian brain or any bayesian scheme in engineering this would be the kalman gain for example is how much gain do i afford prediction errors in updating my estimate of the hidden state that i can't see that place is this this notion of of the the encoding of and the predictions of the optimization of precision center stage in the bayesian brain you see this in many many guises you can see it in the context of multi-sensory integration so how much precision do i

for do i afford visual evidence relative to the auditory um sensory data if i'm in a very very well lit room i might ascribe much more precision to the visual information and that will dominate over my belief updating and my beliefs will be driven by what i see and what i hear but if i reduce the precision of the of that visual information those visual prediction errors and switch off the light so now in the dark room that now becomes very imprecise and now the most precise influences of my posterior beliefs are going to be the the auditory modalities i'm going to listen very very carefully in a dark room i'm going to pay attention to the auditory sensations not the visual sensations so that i've used the word attention deliberately because this from a psychological perspective what we're talking about is deploying the precision in the right way in the right context just is attention it's what we attend to and that attention can operate at any level in the hierarchy can be sort of attention-grabbing stuff in a particular quadrant of my visual field some flutter which is unexplained that can now draw attention at a very low level in the hierarchy or i could be deploying attention to a very very high level of the hierarchy you know in terms of who i'm going to listen to next or who i'm going to place some epistemic trust in what news channel am i going to select next so this process of selecting by assigning or affording precision to this or that this exactly this mental action that we were talking about before it's just deploying attention in the right kind of way and that attention mediated through the precision of either prior or likelihood beliefs has a profound effect on the degree to which i change my mind so it's a really important aspect of this belief updating at the at the neuronal level the predictive coding theory it has two classes of neurons which is the prediction neurons and the prediction error neurons you've mentioned this already but i'm just asking that the neuronal level this is what's happening within the neurons there you have these two classes of neurons that are essentially weeding out is this a prediction and how are we parsing out some of the prediction error this is what's happening at the neuronal level absolutely and it's nice you sort of um ask that question because that allows us to connect um this um predictions of uh precision or predictability um that underwrites optimal belief updating and pharmacology and to the kinds of uh neurochemicals that may be apparent say in things like schizophrenia or indepression or autistic uh autism and um asd and also implicate um all the drugs that have a psycho active aspect from psychedelics through to the kinds of drugs that people prescribe um for say psychotic depression so yeah let's just unpack that so now we've got we've done the the bayes rule and the bayesian brain with in our minds eye free probability distributions or being mixed together in the right kind of way according to their dispersion or their width um but now let's say well how would that be implemented in a in a

bayesian brain well the implementation just is a common filter it's just a predictive coding what would that look like well on one view as you say it would just involve um two kinds of populations one encoding the prediction error and the other encoding the expectations or the hypotheses representations that are generating the predictions that are sent down to form the prediction error so this again speaks to this recurrent message passing between the the prediction neurons sometimes ascribed to the cells that live at a slightly greater depth on the surface of the brain the cortical layers relative or distinct from populations occurring in the prediction error sometimes described as superficial pyramidal cells simply because they live very much closer to the uh to the surface of the brain so you've got this sort of reciprocal exchange of messages where you can think of the deep pyramidal cells of the deeper cells as encoding the position of the posterior let's say some information comes in from the sensory epithelia there is a prior belief encoded by the activity of a deep population of deep cells and they provide a prediction that is now not an apt explanation for this new information that's coming in there is a prediction error the prediction error is reported by the superficial pyramidal cells that is then sent forward to change the deep environmental cells at the next level this is just the belief updating so the uh and now the deep pyramidal cells that were previously encoding the prior expectation expectation here is just use as a word uh to mean the mathematical average or the most likely uh value of our of our belief then the prediction errors can now be seen as doing this changing your mind doing the brief updating just by driving the activity of the depravit cells in proportion to the precision of the prediction error so if you ascribe a lot of newsworthiness a lot of precision to these kinds of prediction errors from this modality in this point in time um then what you are doing is effectively boosting them and you know increasing their potency their gain or their drive so that they cause a bigger belief updating or shift or changing your mind simply because they're driving deep parabola cells towards the likelihood uh to a greater extent and away from the prior than um than had the prediction errors have been assigned a much lower precision so what would precision look like from a a neurophysiologist or a pharmacologist perspective well it's just the excitability of these superficial parallel cells encoding prediction so that tells you immediately anything that changes the excitability or the sensitivity of populations encoding prediction under a predictive coding scheme is going to be physical or biophysical realization of the of this precision which then you ask well what kinds of things could change the excitability or the postsynaptic gain sometimes used in neurophysiology well in the brain there are certain brain chemicals or neurotransmitters whose role is just to modulate the gain of these kinds of

cells and what you then ask them what kinds of neurotransmitters as well ones which many people have heard of um so classic neurotransmitters of a neuromodulatory sort would be things like adrenaline oxytocin a certain oxytocin therefore for a particular reason that dopamine would probably be the uh the sort of the post child of this kind of neurotransmitter well there are others um you know acetylcholine or cholinergic neurotransmission serotonergic neurotransmission of the kind that's influenced by psychedelics and psilocybin for example so you can see immediately if you've heard of any of these drugs you will know that they can have a profound effect on the way you feel or the way that you update your beliefs or respond to changes in sensory contingencies or changes in life circumstances and of course it is these neurotransmitter systems that are responsible for setting the gain and computationally thereby the precision or the ability of prediction errors to drive bleep updating higher in the hierarchy that are used in a therapeutic and also recreational recreational setting yeah it's really important how you're connecting everything here with not only the two classes of neurons but then also bringing in the neurotransmitters and how they're also functioning here as i want to give you a lot of runway here and many of this will start to tie in together so hopefully listeners will be able to see that but give me your detailed explanation and analysis and uh exploration of the free energy principle uh just lay it all out for us just describe what it is how it works how it functions and then how it connects with you know many of the things we've already discussed i'm sure it will come up with active interference predictive coding theory and bayesian brain model so just tell us your free energy principle what it is and how it works and you can be as detailed as you would like i guess i don't want to exhaust you no no it's great i i it's funny because i i feel like at this point we've done all the warm-up so now we can get to the to the to the entree we get to the main course here so tell us about the the free energy principle from uh from uh from yourself right yeah i i have to confess you know that warm-up was probably um a more substantive a pair of teeth than you might imagine we've done we've done 90 of the heavy in the sense that um the free energy principle um provides a first principal account of the of the mechanics that we have been talking about um i'll try and explain what the free energy principle is in a second but perhaps just preface that by noting it plays the same role um as hamilton's principle of least action and you could even argue uh the theories uh that um undergo natural selection evolution in and of themselves they don't do much heavy lifting of an explanatory sort you know for example knowing natural selection is a principle by which you and i are here it doesn't tell me anything about why you've got eyes or a nice little beard or you've got your particular accent it tells me nothing about that right it

just tells me there is a mechanism as to how these things could have emerged so the free energy principle is a little bit like that i'll give you another analogy hammerkill's principle of least action so it's a a generic i would say foundational um um principle that underwrites all known physics arguably and yet it doesn't tell me how on earth i going to throw this ball i can if you tell me all the processes all the mechanics and the initial conditions of a particular instance of bald throwing i can certainly compute the trajectory of the ball but in and of itself it doesn't it just tells you the principles underwrite the actual process of ball throwing i'd actually have to become an expert in ball throwing to actually realize and apply the method of uh or the principle of least action and i use the principle-released action because um the free energy principle is just a principle of least action so it's a principle that could be read as a method that could be applied to a particular instance or circumstance of self-organization or active inference but you're going to have to write down all the actual aspects of that circumstance in order to make the principle do any work and in a sense what we've been talking about in terms of various perspectives afforded by predictive curling the basin brain embodiment um all of these are really um process theories accounts of a realization of the free energy principle so i just put it saying the scene you know and to qualify my statement that the starter was actually the main deal all the hard work and the heavy lifting is really in testing these different process theories and understanding how the free energy principle operates in this particular context with this particular creature this particular institution this particular aspect of self-organization so uh having said that then what is a free energy principle well it sort of transmuted to a certain extent depending upon how the story is told nowadays one would really uh tell the story from the point of view of a physicist trying to understand under a particular constraint and that constraint is that you are trying to understand the the dynamics and the physics and the principles of least action that would apply something now that may sound really trivial um but it's in fact quite an important starting point if you're if you want to talk about something then you have to say well what defines a thing what separates the thing from nothing or from something else and immediately you get into the game of defining how you'd separate the states of a thing from something else the states that were external say to that thing and this leads you into the world of markov blankets and conditional independences statistical devices that enable you to say that there is some independence or disintegrability between the states of this thing and the states of everything else and basically what happens is when you apply hamilton's principle of least action the dynamics joint systems that possess um a markov blanket that separates the thing we usually call it a particle from something else

then what emerges is basically a bayesian brain or the bayesian mechanics that could be applied to a brain but also to a thermostat all to a virus which is why i was saying before the way that we deal with things is in a much uh we usually start you know with very very simple systems so to cut a long story short um what you are doing engaging the same sort of um questions um and formalisms that people in physics would engage if they're trying to understand uh non-equilibrium steady-state dynamics and why is that different from 20th century physics well because most of 20th century physics were all about equilibrium systems it's all about sort of idealized gases it's all about um second laws that apply to closed systems and by closed systems what i mean is that you've got this sort of system of exchangeable particles that's in a heat puff or in a heat reservoir and i use the word heat bath here the question that somebody interested in open systems would be asking where did the heat bath come from and what's outside the heat path and how on earth did the heat valve maintain itself so you now started to ask questions that people in the life sciences have been asking people like um varela in terms of autophagous you know how on earth does the boundary from my perspective the markov blanket the heat buff that contains the idealized internal states of enquiry that define the particle in question of the system in question how on earth is that maintained and even more simply its very existence and persistence what does that imply and if one just follows through the maths that you would normally apply an uncontained set states of the system to a contained set of states that are contained within a markov blanket that comprises sensory and active states and then what you end up with a functional expression or functional form for the dynamics that is formally exactly the same that you would find in the other kinds of mechanics so you know you can think of the free energy principle as just basically introducing a different kind of mechanics that is a complement to on the one hand classical mechanics that would apply to heavenly bodies and massive bodies where there were very few random fluctuations in play because they're so big or cold on the one hand the second kind of mechanics that you get in the absence of a markov blanket would be statistical mechanics thermodynamics uh fluctuation theorems that apply to lots of very hot very small things that don't have conservative dynamics where the random fluctuations dominate over the classical conservative dynamics of newtonian mechanics you can even apply it to quantum mechanics the move here is just to work with the complex root of probability distributions bayesian beliefs the wave function squared now becomes the bayesian belief but you don't have any bayesian nus in either in any of the classical mechanics or the quantum mechanics all the statistical stochastic mechanics and that's because none of these mechanics have a

separation between the inside and the outside and it comes back to something you mentioned earlier on which is the notion of hidden states and you remember and i should have picked up on that when you were framing uh bayesian uh thinking so the whole point of bayesian thinking is there has to be an insight and an so there has to be observable consequences of causes that are unobservable you know before you can even talk about inference there has to be something about which you are making an inference and there has to be some data some observable consequences of these hidden states or hidden causes uh that now you plug into your bayes rule and you know we can then go back to the story about moving a posterior from a prior given the likelihood of your observable sensory data but just standing back from that construct means you now have to have a physics that makes a distinction between stuff on the inside the internal states of the system and stuff that is hidden states that are hidden will now support a bayesian inference and of course this is exactly what is supplied by the markov blanket so the markov blanket is a set of states and intervenes between the inside and the outside and literally surrounds forms a blanket a shield a veil of a statistical sword sort that encompasses and shrouds the internal states so now what plays out on the surface on the blanket states are now the sensory data the observations of sensory evidence that are reporting the influence of the external states on the outside that are hidden from the internal states because they are behind the mark of blanket so you have to have a markov blanket that separates the inside from the outside before you can have a basic mechanics but once you um commit to and follow through the maths that would apply to a partition of a universe into internal external and the intervening blanket states that i repeat can themselves be divided into active and sensory states then you just get the free energy principle and all that says is it looks as if any system that is equipped with a markov blanket and has to be in order to define the thing in question will appear in virtue of the dynamics of its internal states the external states to put that more concretely there it will be an interpretation of the physical internal states as encoding the parameters of beliefs phasing beliefs about external states and by parameters we're just talking about these location parameters and shape parameters parametrize or describe any belief so that's quite remarkable because it gives you a very different kind of take on information because normally when people talk about sort of information theoretic treatments or probabilistic treatments um in self-organization on equilibrium steady state so treatments um that allow for this openness or exchange between the inside and the outside through the markov blanket and they're talking about the information inherent in any particular state you know how surprising would it be to see this gas in this ensemble in this configuration

what we're talking about here is something much more subtle and it's a kind of information that is much closer to a folk psychological or a lay notion of information it's information about things that can't be seen so the internal states now hold amazing beliefs about hidden external states that are hidden from them because of the markov blanket that defines their existence so there's an existential imperative here or one can read this existential imperative this inference process which brings us back to self-evidencing that just to exist as a thing *defi* means that you have a markov blanket which means that there are hidden states which means that if you exist and persist you must be in some kind of technically generalized synchrony with the external states that means that i can read off your internal states and infer that you can as the thing the configuration of external states and this process in a dynamical setting now becomes a process of belief updating becomes a an inference process that can be described technically minimizing a path interval *um* of a in physics called an action which is a time times energy *um* construct *um* as from hamilton's principle at least action in our instance the action is marginal likelihood of the sensory states part of the markov blanket *um* which is approximated by or may in some instances be identical to the free energy so hence the free energy principle is just a principle of least action it's saying that the internal states of anything that exists will have a description in terms of uh technically a functional of beliefs about external states so that was a very brief so so let me let me try and let me try and summarize here if i if i can get this so what you're saying is that all biological self organizing systems they are ergotic they have the markov blankets they exhibit active inference and their self for self preservation or to put in another way all of the quantities in a self-organizing system that can change will change to minimize free energy do i have this right yep okay so how so i want to i want to go to this idea of hidden states right so you obviously talked about equilibrium and i think maybe i would determine in my world with something like homeostasis right you know people in psychology or people that do physiology they like to talk about homeostasis so maybe you can chat a little bit more about the role of homeostasis or equilibrium in preferred versus expected uh various physical states i do want to introduce a little bit because i know we've mentioned them a few times you know mark solms i know you guys have worked together i've talked to him he's fabulous *um* so he has a psychoanalytic angle here which i'm very fond of as the impact of the freudian model of of drive energy so maybe this is i'm adding something here so maybe you want to talk about homeostasis but my question about freud is how does his conception of drives or drive energy how is that consistent with your free energy two really easy questions right so you want to go from bernard to freud so but that was you know that that was i think a

clever move to to introduce the notion of homeostasis so what we're talking about here is just a generalized homeostasis so you know um you can regard um this non-equilibrium steady state you use the word ergodic which was interesting um we try not to use that word because that brings all sorts of baggage what we're implying um in terms of a non-equilibrium steady state is over a limited period of time realizing that of course at a slower time scale things will evolve so we don't imply that the system ever attains a steady state it's just that it has a steady state solution to its dynamics but that solution may disappear after a short period of time but within a particular time scale um what we are talking about is um the fact that it has um maintained its markov blanket which means that it must have established some kind of synchrony or balance between the inside and the outside that we can now articulate as a principle of least action or a principle of least free energy or maximum model and then the question is well is this consistent with other theories and other accounts of self-organization or global function of a system and the answer is yes absolutely and so your homeostasis is one of my favorite accounts but there are other accounts you can also say well this um fundamental objective function that appears to be minimized remember this is a principle of least action it looks as though the system it looks as if the system is trying to minimize something so that could be a cost function which could be a negative reward function or utility function so you can now spin off all of reinforcement learning and economics in terms of utility theory as one expression with a particular commitment to you know understanding the mechanics and the you know the rhetoric of that field either in the economics or psychology sciences um but it's just really an expression of that there is something to be optimized or it looks as if the system is optimizing something because it conforms to this principle of least action where action is the negative log evidence of the free energy all of reinforcement learning and utility theory is just a statement i think i can describe the behavior of this system as an optimization process what i need to do is to identify the function that's been optimized and of course from our perspective that's going to be or minimized in our in our context that will be a free energy it's converse the marginal likelihood and the evidence of my model the world so you can rewrite all of reinforcement learning and utility theory as basically an expression of or an optimization process of self-evidencing where you now start to interpret the prior beliefs as um or the logarithm of the prime beliefs as a reward functional preferences so just because um i am sitting in this configuration with my markov blanket uh means that the if i now interpret my behavior in terms of basic mechanics i necessarily now have to understand it or write it down in terms of a prior belief and a posterior belief in the updating and it turns out that that prior

belief just is the distribution of the um the steady state solution that we were talking about previously which means that just existing is a statement of your preferred states of being and you can articulate that mathematically in many different ways so you know for example in the theory of random dynamical systems um the set to which you will always return will be uh called a pullback attractor or a global random dynamical attractor these are just the states which i'm attracted these are my preferred states and because that's what's been optimized from the point of view of reinforcement learning or utility theory you can move them to the observation that the the negative of these utility functions or value functions is surprise that's also known as self-information information theory it's just the prediction error we were talking about before we try to minimize that and then from that you can read up all sorts of information theoretic treatments of optimal behavior such as maximum mutual information theory infomax theory you could even go as far to a certain extent as integrated information theory if you wanted to but information theory accounts that posit some um some metrics some functional information theory measure that is uh extremized of course the free energy principle is just one of these it says the information theoretic measure is just the free energy which is a bound on the negative log model evidence so marginal likelihood but then you come to your set of examples which would be um more from the point of view of theories of either of a sort of physical sort like cybernetics and then you get into things like the good regulator theorem and um uh or synergetics which are entirely consistent with this sort of optimizations of your hamiltonian principle at least action um but if you're a physiologist all you're saying is just hemistasis this is just a generalized home mistake it's um trying to place bounds upon or it looks as if you are we're in minimizing your prediction errors if you now think of a particular kind of prediction error as the prediction error any autonomic physiological variable and your a priori set point or belief about what it should be in minimizing the prediction error you are doing homeostasis so minimizing that surprise in the particular setting of the surprise being about physiological states just is homeostasis and if you just generalize that you're all you're doing when you're describing a particle that keeps its states within particular bowels within its mark of blanket it necessarily has to resist random fluctuations and gather itself up to convince a generalized uh generalized and then the final interpretation of course is just in terms of self-evidencing and the bayesian brain hypothesis but these are just different perspectives on the same underlying underlying dynamic so having established the the privacy of uh homeostasis in its uh full and generalized sense we just have to put freud into the story which is not that difficult actually um and i say that because i should

acknowledge that mark zones is the world expert on freud and you know i don't know if he spoke about his books on on all the translations uh so my understanding is childish and unscholarly to say the least but from what i understand having spoken to people like mark many of freud's um foundational ideas were coincident or shortly post stated much of the work of helmholtz so when when what one might easily imagine that freud thinking about sort of um his energy constructs um in terms of uh you know a freudian he probably had in mind the same kind of thing that helmholtz had in mind and one i hadn't seen anything written or i haven't read about this but certainly from secondary and tertiary sources and the helmholtz did have a a notion of a neuronal free energy or a neuronal energy that was the thing that could be optimized or minimized in order to explain unconscious inference he never really articulated or found it um but he certainly had the quest in mind that i fondly or would now submit would be the variation free energy it fills everything that he wanted and is entirely consistent you know with his more formal um formulations of of perception and also much of physics um so my guess is that the basically freud was was um taking that notion of um an optimization in terms of uh an energy function as it was articulated and conceived of at the time that these are the inception of these ideas that you're bringing canteen philosophy into physics but without the machinery afforded by richard feynman to deal with it with your in in a simple and graceful way in terms of mathematical equations uh and without of course the uh you know high performance computing that machine learning subsequently enjoyed was able to demonstrate proof of principle that you know this is the way to format the problem so you know fry probably got as far as he could in the absence of fiberinesque maths and pathogen formulations and big computers but pursuing these kind the same kinds of ideas so it needs of no surprise to me when mark tells me that there is a fundamental consilience between freudian formulations and those that you would find that inherit from uh feynman and you know and all the other contributors to these you know to this formulation of sense making um so i i would imagine that freud gets into the game quite gracefully um you know the particular point i think that mark zones would emphasize here though brings us back to this um this metacognition and beliefs about beliefs and in particular the ability of certain hierarchically abstracted sophisticated particles like you and me to actually physically encode and manipulate uh and do their bayesian leaf updating on beliefs about beliefs and specifically the precision the uncertainty to know our own uncertainty to know our own minds and the deployment of that in terms of overarching homeostasis this extremization of minimization of prediction error or free energy or maximization of marginal likelihood or moral evidence and underwrites that

kind of mental action and may on mark's view at least um be the essential quality of any um any fur any feeling or any sentience so he has a lovely phrase called felt uncertainty in the same way that we um visually palpate the work that you know the visual scene with our eyes and what he's suggesting well the the way that i interpret his his his poetry and he may be very cross if he hears this but the way i the way i interpret it is that you can palpate your your own uncertainty palpate you can feel your own precision and furthermore you know and have prior beliefs about the way that that uncertainty should be deployed or changed and you update your beliefs about that precision and the mechanisms that do that um may well be um those that are at the heart of the feelings under right quality of experience of the kind that the mark would ascribe to you know to consciousness felt uncertainty yeah it's it's interesting with freud because obviously a lot of it in terms of the the the drives was this um uh i'll say is it it was a in an interesting way i mean the way he framed it was in terms of the psychological hierarchy that he was trying to give as a model was this element of tension reduction rooted in our our biological and evolutionary nature if you were to make a comparative analysis here you would say that that was probably a passive approach right because it's always something is happening how do we how do we resist how do we how do we get rid of the tension reduction is it in a way it kind of pass away as opposed to maybe say the you know philosophy of frederick nietzsche who had a more active way of trying to understand energy and drives and power and things of that nature okay so some of the further questions i want to ask you are since we've done a a very nice or you've done a very nice job of of explaining the the free energy principle i would i want to ask about how it applies and then a little bit of some criticism so i guess the one connection here is about consciousness so as we've been alluding to people many people have ideas about consciousness where resides how it functions so everyone from um chalmers to dimasio to obviously mark has has made some pretty um extraordinary uh research on this in terms of finding it in the uh in the mid brain the brain stem which is fascinating stuff there um and all and now seth i mean there's a bunch of people that have been talking about consciousness it's sort of trendy how do you based on everything we've discussed um bayesian brain free energy principle how do how do we can this tell us anything about consciousness more accurately or more clearly about what it could be where it could be how it functions um what is i guess your your kind of a stamp on on this topic in terms of i think the first thing to say um which is often said by philosophers is that the free energy principle is not a theory of consciousness that sounds um slightly deflationary but in fact it's quite a useful thing because if it's not it wasn't designed to explain consciousness and um certainly people like me have no

pretensions to try and get into that game but it will be very interesting to see if it holds the formalism that allows people to address certain hypotheses or questions about consciousness so that that is deeply engaging and just for your interest um there's currently just been funded a um john templeton foundation collaborative adversarial collaboration to play off uh active inference against information uh integrated information theory so there are lots of academics and theoreticians out there who are sort of um interested in what more formal accounts of sentient behavior will bring to the table in terms of answering questions about consciousness um so everything i am um saying is contextualized by the fact that you know the french principle is not there to explain consciousness if you had to answer some of the deeper or more challenging questions um such as um david chalmers hard question and or accommodate um some fundamental dialectics you know cartesian dualism for example what space is on offer um from the free energy principle and i think there are some quite clear answers there and we can take this from a number of directions um i'm gonna i'll take two because they follow on from from um some of your uh pressing questions about sort of drives you know from the freudian perspective um so what what did what would drives look like um from the point of view of the free energy principle but let's not talk about the free energy but let's talk about active inference as an actual process theory for you and me um that are if you like uh conforming to the free energy principle and so the the first thing that um one has to note is something we've spoken about before that we have a calculus four beliefs about something that immediately introduces um a duality in terms of interpreting internal brain we can either talk about the problem probabilistic population dynamics and the information populations on their activity which would be um uh intrinsic to the brain you know how likely is it that my neurons are going to fire in with this particular pattern of this particular part of the brain in this particular circumstance which of course is your one of the key questions that we address with your imaging you can under the free energy principle or active imprints say well hang on a second there's another kind of probability structure a belief space because we've just said for every pattern of neural activity there is encoded an implicit representation of beliefs about the now what that means is technically that not only is there um a probabilistic description of my neural activity but that neural activity is equipped with something called a statistical manifold by by which i simply mean that every point on this manifold that corresponds to a pattern of brain activity is equipped with a belief structure a belief distribution a bayesian belief about something else about the outside about you the causes of my sensations and that means that i can now talk about an information geometry of a different sort which is now about the outside

and extrinsic information geometry and that's quite a nice picture this notion of belief updating we've been talking about phase in belief corresponds to a movement on the statistical manifold so you imagine that every pattern of activity in your brain corresponds to a point on this abstract manifold this surface is a very high dimensional space and every time the pattern of activity changes in virtue of your neural dynamics you make a movement on that space and that movement is belief updating it is exactly the same as moving from the prior to the posterior that we were talking about before so sensations come in and they move point on the statistical manifold to produce the belief updating so you are literally now describing mathematically in with an information geometry belief updating and the process of imprints the the key link here is that the the measures the the metrics that define this information geometry of belief updating something called fisher information metrics and mathematically they correspond exactly to the precision we were talking about before so technically it's the curvature of that space as measured by the uncertainty or the precision of your belief updating so there's an intimate relationship now between the precision that we were talking about in relation um say mark zones and just to slip in here that the cells of origin of those neurotransmitter systems that control the post-synaptic game that mediate the precision are exactly in the mid-brain structures that mark thinks is the seat of consciousness so this is all entirely internally consistent but it's also consistent with the fourier notions of drives because you can think about this movement on the this belief updating has been driven by forces what are these forces they are the prediction errors they are the um the surprises they are the effect of the likelihood on priors that force it or move it to produce the posteriors so now we have this notion very elemental notion of um a drive probably sort of oversimplified from a freudian perspective but i think the mechanics so holds true here that the drive is literally the thing the the uh role of the prediction errors in driving belief updating from prior to posterior police but crucially we now understand this force this drive as being gated or nuanced or contextualized by the precision which brings us again back to the fundamental importance of getting some sort of metacognitive beliefs about beliefs or predictions of uh of precision right in terms of enabling those particular forces and drives to be manifest in the way that we both organize our neural activity but also um uh explore our belief space by moving around on the statistical manifold so you can write this down very precisely in terms of information geometry there's lots of really interesting maths that gets right into the world of gauge theories and the like all telling the same story that the the the potency of these drives depends upon the curvature and the curvature depends upon the precision or our our beliefs about the uncertainty

the point here though from uh from the sort of consciousness point of view is that there are two descriptions of movement on this statistical manifold there's a description from the extrinsic perspective in terms of the probability distributions encoded by a point on the manifold and there's another intrinsic description which is a simpler one just about the probability of being having this pattern of your activity so there are people that are thinking uh about uh colleagues such as uh vanilla vice together have um put this forward as a way of repairing dualism you've basically got a dialectic for free mathematically in the information jointly ascribes particular physical instantiation in terms of neuronal activity and patterns of neural uh neural firing to any point on the statistical manifold that is accompanied by beliefs about something else which would be the sort of the more mindful aspect of a cartesian divide so you've got this sort a molistic account that has these dual aspects of mindful belief updating and the physical embodiment of that um i think it would be a potentially useful way to talk about sort of dualism versus versus uh modernism when it comes to you know some some possibilities to be historic but to my um limited understanding sort of uh you know foundational issues in uh in consciousness research and that that that would be that would be sort of um one um if you like opportunity for the the bayesian mechanics afforded by active inference and the free energy principle produce some quite clear predictions um about the relationship between mindful stuff and brainful stuff um where there has to be if you like a convergence within this within this information geometry the other thing it brings to the table is that it is the movement on this manifold it is moving through a belief space that is the essence of inference so if you remember the free energy principle is just trying to describe the dynamics a system that has a non-equilibrium steady state solution for a certain amount of time inherent in the use of the word dynamics is you're looking at a process that is unfolding in time what that tells you is that if you want to talk about consciousness then it may be best to think of consciousness as this process of uh of belief updating in exactly the same spirit that we talk about evolution as an evolutionary we don't talk about the state of evolution at any particular time that's divorced from the dynamics the paths and the process itself the the process of evolutionary selection um as opposed to say selecting a plan or a particular belief or a pro you know um you know are updating my beliefs in terms of the process of uh inference under the basic brain hypothesis you know we talk about the process as so what that would mean if you just translate that perspective which is inherently dynamical to consciousness research is that questions about what it is to be consciousness have to be recast in terms of what processes must you convince in order to qualify as having a conscious process so states

of consciousness and levels of consciousness are um only meaningful when they apply to something that is changing in time essentially moving on this on the statistical manifold so that's uh from my point of view a couple of things that are brought to the table by having a formal mechanics or calculus underneath belief updating to have a calculus of sentience a physics of sentience that tells you immediately you're talking about processes and of course if you think of you know in terms any optimization or casting things in terms of any optimization um then you are talking about the process of optimization and indeed even um from the point of view of the the semantics of things like variational principles and least action you know the least the minimizations aspect of it talks again to a process that is you know playing out over time so that that would certainly be um if you like one's constraint on author for anybody who talks about consciousness you know are they committed to a way of thinking about consciousness that just doesn't have any um any meaning in the sense that it's it's about a particular state of being um or can they recast or formulate their hypothesis in terms of uh of of a the third level um or the third sort of um thing on offer um with active imprints brings us back again to the generative model so remember previously we were talking about differences between uh thermostats flies mice and men i was very comfortable talking about those differences uh from a unique perspective which was simply what are the differences in the generative models so if you can boil everything down to the for which you are trying to secure which is the generative model then you can start to talk about different kinds of consciousness in a graded way in terms of different kinds of genitive models and we've already had this conversation in the sense that you know i can have a genetic model for a thermostat which has um which is temporarily very thin it has just you know very very limited temporal horizon and then we graduate to particles and creatures that have a more allostatic as opposed to a homeostatic um way of engaging actively with their environment that require certain temporal depth and then we move on to um the the opportunities when you now have a charity model in the future for different outcomes different counterfactuals and then we move into anil seth's notion of counterfactual richness but noting the counterfactual richness as one way of understanding consciousness or conscious process can own is only allowable when your charity model has a uh the capacity to model counterfactual outcomes in the future um and then we get to um the remarkable counterfactual it comes under write things like this you know the philosophical zombie and braided of that and then we start to talk about you know we can go from mice to men to philosophers and then start to engage uh engage with that kind of conversation so that's another you know sort of structural architectural formal

thing uh or um way of dealing with these questions that you get from the physics and the mathematics which wouldn't otherwise be there if you're just using narrative arguments historical arguments i was going to say you were talking about this convergence and as you were describing that i was thinking about how there is a kind of multi-dimensionality or a continuum aspect of this having a being a process of things as opposed to just this static kind of state which i think based on many of the things i've read on consciousness that makes a lot of intuitive sense and i think the way that you're talking about this underlying calculus to try and describe a kind of process makes a lot of sense artificial intelligence so maybe we can make the the leap a little even further right mice to men to philosophers to ai right how do obviously there's there's it's sort of connected with the consciousness question although you could make artificial intelligence that doesn't need to be conscious um i don't think it's a prerequisite although some people may argue about this but in terms of again the this underlying calculus that is there with the free energy principle what is it i guess specifically or something that gives it this uniqueness that can give us information about artificial intelligence obviously artificial intelligence is a very broad categorization so i mean my you know dishwasher and microwave is a form of artificial intelligence right at a very very low level um and then there's artificial intelligence that we see in science fiction uh stories uh such as you know humanoid robots that are sentient and they maybe have feelings and you know things like that over time so there's a whole gamut here and of course you could make the the offshoot of well first you have to define intelligence which is a very long footnote which we don't have to do although if you want to i would totally be fine with you explain that as well but just how do we understand the elements of free energy principle for artificial intelligence now and how it could have potential utility as we continue to grow and expand our mechanics or computational aspects of part of your question was you know how how does a free energy principle give you the opportunity to get more information about the uh the nature of intelligence and as you uh integrate sort of artificial intelligence um interestingly that that that getting information about is actually at the heart of the answer i'm going to give you so the answer i'm going to give you i'm going to foreground though um one key thing which is um in response to a previous question here what's the free energy principle for um very much like the theory of natural selection it doesn't really do anything afraid because it's it's it's very pretty and you can have philosophical debates about it which i'm sure we'll come to um but you know why is it useful and the answer is um if you read the free energy principle as mechanics that is apt to describe systems that have solutions to their non-equilibrium steady states in

terms of an inference process then the answer is it's pretty you know it is not useful however once you know the functional form of the equations of motion the dynamics the updating that have to be in play in order for the system to be intelligent or to be to show a self-maintaining or autopoietic-like behavior what you can do write into an artifact the generative model and all the prior preferences entailed by that generative model to simulate an artifact that has the same properties of itself maintaining and maintaining its markov blanket why would you want to do that well i think there are two cardinal reasons why you might want to do that why would you like to build a markov blanket maintaining building machine that's doing inference that it basically is realizing principle of least action that prescribes the differential equations that you would write into a computer simulation or indeed a robot or a you an artifact that was supposedly showing artificial intelligence well i think there are two reasons to do that one you might want to build an active inference agent an artifact that's actually doing active inference how would that differ from artificial intelligence as we currently know it that's a really good question there are very clear answers and i'll and i'll give you those answers in a second but just to conclude this part of the answer uh with the second motivation that you might want to build synthetic agents with synthetic sentience under the variational principle of least action that is the free energy principle is you might want to make inferences about somebody else and i know now thinking from the point of view the psychiatrist you might now want to try and simulate choice behavior and under a particular model and then adjust the preferences and the price of that general model until you reproduce the kind of observable behaviors that this patient has or this particular cohort shows so what we're talking about here is essentially using simulation models as observation models of real world people you know and other creatures in order to try and understand their internal dynamics and their internal there's something quite fundamental um about that particular take because remember before we we talked about the markov blanket in shrouding the internal states of something from the external states so that there is if you like an impenetrability there's a fundamental statistical divorce between um the internal states and the external states meaning that the external states are hidden from the internal states but the converse is also true which means that from the outside the internal states of any creature or person or loved one are forever hidden from you from the outside so you can only ever infer the machinations and the dynamics and the belief updating on the inside which is interesting because that means you much of my day job in your imaging is trying to peek behind the markov blanket to see what's going on but you you can't do it properly your otherwise you'd have to break you'd have to reach the markov blanket to

become invasive or study dead things in order to try and infer what was going on so that means that you need to use inference just to understand the internal dynamics of the brain of a person which means that the only way you can do it is to have a genetic model of a genetic model which means that now you've got another use case for active influence in the free energy principle which is building models of people and patients and then adjusting them until you use the adjusted model that renders the um the behavior of the model um very similar to the behavior of the patient to infer this is how the patient works or at least this is as if you know and as if explanation are counted as patient so those are my two principal motivations to use these as a principled model of choice behavior on your rural responses that i can measure empirically to answer my hypotheses about how the brain works there is another agenda which we've touched upon which is you know more the fun of creating artificial general intelligence of general general intelligence or agi and the distinction between artificial intelligence and agi or the next wave of um you know artificially or intelligent artifacts um i personally do think um is staring you in the face um um in terms of what the difference is between active inference um under more sophisticated genitive models that have the ability to model the consequences of action if the end point of this argument if it is the case that i have to minimize my prediction errors or free energy or at least look as if i am minimizing my prediction errors or free energy just to exist if it is an existential imperative for existence then a priori i must also believe that every action that i choose every choice that i make must be that which is the most likely to minimize prediction errors or free after i've made that action the consequences of my action are now scored in terms of their ability to reduce the free energy i expect following an so then what you can do mathematically is look at the expected free energy and ask how could i interpret that what does it look like what are the qualities how what are the first principle accounts of a good choice or a good and when one does that um it looks like as you might expect um there are a number of different interpretations of this expected free energy which is a big thing that active influence brings to the table so this is an objective function that now replaces utility functions and reinforcement learning it also replaces the um principles that underlie active learning and things like optimal bayesian design in the sense that it grandfathers these things and puts both of those two aspects of good behavior um together and in putting them together this objective function that underwrites the right choices the right behaviors is just expected surprise or uncertainty which means that all i need to do is to choose those behaviors that minimize or resolve uncertainty or maximize information gain under prior preferences inherent in my generative model so that last

statement is quite crucial and what what it transpires um to be also the expected free energy transpires to be a mixture two components one of which is an epistemic information hungry seeking component and the other one is a more pragmatic preference seeking component and put them together they are just then minimizing overall uncertainty or minimizing the expected expected free energy crucially the epistemic information component is the the thing that underwrites curiosity so if you look at the artificial curiosity literature they um the the key problem is has been um articulate and mathematically formalize the drives to resolve curiosity the sensation seeking the inquisitiveness that characterizes all the you know all of us certainly in certainly in the context of this conversation but uh you know i think all of us as as little scientists i just need to live we have to be curious creatures and the expected free energy is basically that curiosity that intrinsic motivation that's another word in your neural robotics developmental neurobotics also known as expected salience and it puts it together with the surprise minimizing aspects endowed by having prior preferences in your generative model i i expect myself to do this kind of thing or encounter these kinds of outcomes but uh because i'm um contextualizing that with this information curiosity seeking component i've now got a complete objective function that essentially equips active inference agents with curiosity so that is what is missing in ai that no alexa or no iphone is ever going to ask you a question because it's curious about you so the next generation of artificial intelligence if it complies with the you know the variational principles of least action as articulated under active inference should show a genuine interest in you because it will act or it will select those questions and those actions or look at these websites or these engage in this kind of data mining simply to resolve uncertainty about its generative model of the world and if instead of model of the world um contains you it'll want to know about you and it will ask the best kinds of questions that will maximally resolve its uncertainty about you so it'll be genuinely interested in you um and you know in that sense then i think because it's got this um inherent in its selection of its actions i think it will then show a kind of artificial intelligence that is much closer to the generalized artificial intelligence that people have been been aspiring to but it does rest upon this upon putting the curiosity in as part of the package by minimizing the i i have this this chilling a fantasy in my mind as you're telling me this about you know my my uh alexa or my my siri on my phone asking me questions about just a very very chilling kind of thing about it but i do agree with you i do think that that's probably the the next step the next place because of how good algorithms are at trying to predict based on things that we're searching or looking or things like that you know it's trying to to to to accurately get

something if i put a search in google now it takes you know not even two seconds right you know 10 years ago it took longer and you have to find all these things it's trying to curate the things for you that you're searching i could imagine that there's more of a not just that the the ai is is receptive but then expressive in what they're trying to ask and inquire of you the the person that's searching which is uh there's a little bit more you're right there's like a more of a two-way street there which is definitely uh it's a little bit of a scary thought to think about i think there could be really cool utilities to it but i definitely think there's obviously like many things in inherent danger before i want to ask you about the imaging techniques you've done because that's i mean as you say your day job is is uh that's what you do and obviously you've been prolific in those those areas i just want to ask one thing just because it's relevant now and i think as we continue uh moving throughout this you know second uh decade i guess of the 21st century um it's about your work with copen19 and i know this is a big big area big topic and a lot of noise about the pandemic in all aspects i just want to ask you just generally you can briefly just summarize you know kind of how what what your work was i believe was with uncertainty and predictive models and how we understand things and how we understand um certain types of immunity and things like that just kind of how you became involved or how you how you think about about you know the the pandemic and and covet and how how you've been doing work in in that area yes of course so um the answer to that question does come from my day job which is using brain imaging and empirical data acquired through a variety of different ways of peeking or breaching the mark of blanket or looking inside people's heads so you can do that with things like functional magnetic resonance imaging which gives you sort of whole brain images every second or so in reports fluctuations in coarse-grained neural activity in different parts of the brain right things like magnetoencephalography equivalent eeg procedures which give you a much temporary resolved picture or signals reporting electromagnetic fluctuations in your activity over your fractions of a millisecond but the price paid is that you you lose the spatial resolution but the point being is that these are ways of looking at the brain in action and um affording data empirical evidence for various hypotheses about the underlying functional brain architectures of which we have been talking a lot so the hierarchical architectures the recurrent message passing entailed by different say predictive coding or other theories and in particular um something from my perspective um failures of those architectures in terms of false inference uh that ensues from for example breaking the precision control that we've been talking about and how that might show up in people for example with schizophrenia or depression or other

neurological or psychiatric conditions so that that's my day job and link to the covered research is quite um easy to uh demonstrate in the sense that what you are talking about when looking at a message passing architecture and your role or message passing architecture that is doing the belief updating that is um enacting the drives afforded by prediction errors in doing the belief updating and causing activity in different parts of the brain this obviously can also be expressed as a communication and a dissemination of information or influences so you've got the spread of messages um bits of information from one neural population to another neural population in order to make sense of the brain imaging data you actually have to have an underlying genetic model of this process and now we're talking about a genetic model of the brain and i as a scientist are using to make sense of my scientific or empirical data that comes from the brain imaging and and i've deliberately phrased the the fundamentals of these generative models in terms of spreading information among neural populations because of course that is exactly the same kind of genetic model you'd use when trying to model the spread of a virus amongst the populations that constitute a country or a region or a city so my involvement um in the um the modeling of the coronavirus epidemic comes straight from the observation that the generative models the state-space models the convolution models these are just technical terms and the kinds of models we use to make sense of brain imaging time series generated by populations of neurons could be very quickly and efficiently repurposed to make sense of the epidemiological time series generated by populations of people that were in various states of infection symptomatic you know vaccination and the like so for the past two years my weekend job has now been split behind something called bayesian model selection or structure learning which is basically fiddling with your geometric model until it minimizes its free energy or maximizes its marginal likelihood in relation to the uh the data that's incrementally becomes available and different kinds of data um on the one hand and then trying to do the um the free energy principle and make it simpler and more and more accessible on the on the other hand so that's how i became involved you know different levels of engagement working um as a panelist on the independent stage which was a sort of a public facing version was meant to be a compliment a public-facing complement to the uh the scientific advisory group on emergencies that were supplying the cabinet office and the government with advice we were much more in its original formulation sort of public service facing information um uh dissemination and public engagement sort initiative uh run by academics um i have been involved on other advisory bodies um largely motivated or resting upon um the predictions made by um what's called dynamic causal modeling of the time series

data that allows you to optimize the parameters of that generative model given all the data you have up until this point in time and then run the model forward under you sometimes in different scenarios that allows you to make sort of forecasts about what would happen if i did this or did that it can be useful um nuancing the advice given to to your politicians and decision makers when considering you know should we relax lockdown rules at this point should we encourage this age group to take up vaccines for example yeah no that's that's super super salient and instrumental so that's that's uh that's incredible work um so we've already kind of talked about it so let's let's get into your day job your your with the imaging techniques um the thing that i think is your initial claim to fame here is the statistical parametric mapping and then i think also subsequently the voxel-based morphometry these are the two imaging techniques most people will know a ct scan an mri scan i do want to ask you about fmris some people may know about diffusion tensor imaging there's all different types of ways of scanning the brain uh so maybe just tell us what these two things are the statistical parametric mapper and the voxels and then just kind of how that sits in with a mapping of the brain uh uh overall right well you introduce the notion of a ct scan and then many people will also be familiar with x-rays your images um that peep behind the curve and get on you know give you a picture of what's going on on the inside um so statistical parametric mapping can be conceived of as producing x-rays of significance of significant activation and by activation i mean a change so going back to this notion that the interesting stuff is in the processes the uh from the point of view of the free energy principle the belief updating so you needed a way of forming x-rays of a certain kind that reported where the belief updating and the neural activity and the hemodynamic or the blood flow consequences of that human dynamic activity where and when they were occurring um in the brain so spm or cystical parametric mapping is just that it's just it's just the construction of x-rays or ct scans where the values of the image at any point in the image usually known as a pixel or a voxel correspond to a statistic testing a particular hypothesis about the pattern of the um the fluctuations in your activity over which you have usually some experimental control so when people talk about activations what they basically mean is has there been uh have i been able to evoke neuronal or belief updating and neural message passing and thereby neural activity and everything that supports that neural activity by presenting this kind of stimulus to the subject relative to that kind of stimulus so you're just comparing the evoked responses over time and then summarizing those differences in terms of a statistic and then putting that statistic for that part of the brain into an image and thereby creating a whole brain map or a you know statistical x-ray of the functional anatomy

so it's now unlike a ct scan which is a structural anatomy this is not the functional anatomy because you're you're if you're like summarizing a functional process and very similar approaches um you are taken to um in eeg or electromagnetic and electrophysiological um procedures you mentioned voxel-based morphology that is simply the application of statistical x-ray this spm like technology to the special case of sequences of um structural images so you're normally in the game of trying to this group of patients say with alzheimer's disease or parkinson's disease do they show a systematic difference in their structural anatomy again in this instance using a an spm or a statistical x-ray but in this instance the statistic is reporting the significance of the difference in the composition of the anatomy as measured with um with magnetic resonance imaging usually or could be ct scans um by uh but using here not time series data but usually data points from multiple subjects that can carefully uh carefully selected so it sounds like there's a there's a there's an integration of trying to have the integration of looking at the functional pieces and time of a process for a structural picture is is what it sounds like it's trying to do how you use this for schizophrenia um particularly how did you use these methods to understand more schizophrenia and how literally the images of the brain for for individuals with schizophrenia how it looks and how it functions and maybe just talk about that as well i'll just as a as a as a footnote i worked in patient uh with individual and outpatient but in-patient with individuals that have schizophrenia so i i've uh i've been around these folks for good part of my career and um they're they're very very fascinating um individuals in terms of personality in terms of their cognition in terms of their brain functioning and you know being on plenty of uh you know treatment plans and big big big long meetings with psychiatrists and neuroscientists uh trying to understand schizophrenia but then also understand and how to try to provide clinical services to individuals with schizophrenia so i'm curious as well personally almost of of how how did you use these these techniques to try and understand specifically uh individuals with schizophrenia right again that's an excellent question and it speaks to the motivation for much of the theory of the work that we've been talking about so understanding um what has gone wrong in certain people um qualifying this of course the you so the schizophrenia would not like you to think that their brains have gone wrong that's right that's right right they just have a different reality that's all exactly um i felt that's mathematically true by the complete class there but that's another story um the way that all this fits together is in terms of um understanding much of psychiatry and many neurological syndromes in terms of false inference and if you can understand the experiences of a psychiatric patient in terms of false inference then now we already have a

mechanics that is what we've been talking about active inference and free energy principle and predictive coding basing we already have a mechanics at hand to ask at what point has there been some aberrant um message passing or belief updating that could lead to this uh this false inference so you may be asking well why on earth would you want to cast all of psychiatry is false inference well it's a very natural way of accommodating that that's where the symptoms lie that's where the ontology of psychiatry lies it's in belief updating it's in beliefs about the world so when i talk about false inference i i mean it in a you know in a simple sense of type one and type two errors false positives and false negatives so when you think about say a hallucination what is that well it's just inferring something is there when it's not it's just a title on a kind of what about neglect syndromes agnosia dissociative syndromes um they are just false inferences of the second kind if something is not there when it is uh and i think you know if you think about it nearly all psychiatric conditions can be construed in this way either falsely unfair something is there when it's not and vice versa so what that means is you you now move on to the uh the more challenging and pragmatic questions well what could account for kind of um false inference and the questions then can be um either to the structure of the implicit generative model that that is in tamed uh so entailed our brains or the the process theories that undergird the message passing and the physiology of that message passing and what you're basically talking about is a form of disconnection now this disconnection may sound odd to talk you know immediately about disconnection but remember the architecture of any distributed system is defined by its connectivity the very notion of a hierarchy yeah depends upon the absence of certain connections that would otherwise transcend levels in say a linear subsumption hierarchy so connections are everything in terms of sculpting and defining the architectures endow the brain with an implicit generative model of you know its lived world um and furthermore the message passing you know enacts the drives the belief updates um you know the precision weighted prediction as we've been talking about and those are just expressions of influences from one part of the brain to the other part of the brain so it's all about the connectivity and that means that you need to now um use the empirical data in a way that enables you to answer questions about disconnections and they can come in many flavors you as a neurologist you know subscribing to uh early ideas of um disconnections from people like gashwind um in terms of lesions the organs of connection in the brain the white matter tracks that connect the other long range fibers that connect different parts of the brain these can be sort of macroscopic disconnections from the point of view of schizophrenia research that that idea was first proposed by wernicke who was famous for you know his

studies of specialization functional specialization in the language system but his big gift uh in terms of psychiatry was was the junction hypothesis that you could dementia precocks are early formulations of schizophrenia in terms of disruption to these uh the organs of connectivity the white matter tracks that we currently um that we currently study with things like um diffusion uh weighted resonance imaging or diffus tractorgraphy then based upon that uh that kind of imaging but there was another perspective which most people ascribed to boiler um which spoke more about the sort of the fine-grained disintegration of the psyche the the loss of coherence and um you know literally you're a dissolution uh at the level of mental life so from our perspective that would be belief one might nowadays understand much more as a microscopic disconnection the level of synapses the little specializations little buttons that or plugs that connect one neuron to another neuron at the end of the the axon or the long fibers that emanate from a particular neuronal cell there's a a whole bunch of pre and post-synaptic specializations called you know called synapses which you basically in the little plugs at which all the neurotransmitters act um so clinical neuroscience you know a failure of or a disconnection or a failure of uh integration at the synaptic level will be known as a synaptopathy a pathology of synaptic connections so that's the kind of disconnection that most people have now in mind when thinking about the abnormal message passing that might um produce the false inferences characteristic of different psychiatric conditions that means you need to be able to measure this you need to have an in vivo in life way of measuring the synaptic efficacy the connection strength the gain that we were talking about before that is modulated through the precision engineered message passing getting the excitability right these are all different ways of expressing the synaptic efficacy the strength of connections in the brain and if you think that things like schizophrenia are manifestations of false inferences that inherit from a very pernicious and particular kind of psychopathy precludes getting the precision right getting the excitability of say for example neurons reporting prediction errors correct then you need to be able to measure it and that's where the third technical um contribution to brain imaging comes in which is the dynamic cause of modeling that that was used to study the the pandemic so we use the the the the dynamic causal modeling to basically um study the spread of viruses from person to person um but it was originally devised to study the synaptic efficacies that underwrite the spread of signals from one member of a neuronal population to another one so that's where it's at at the moment i mean it's difficult for me to um summarize you know what we were doing and what we are doing um in a nutshell because a bit like the your your first questions about how much we i have achieved in my career i think

this field also it's a long journey and it's very incremental and sometimes you're quite surprised by how far you've moved but if you look about the progress from this year to last year it's almost zero you have to look over 20 years from my perspective at least dynamic cause of modeling that was that was explicitly um done in order to understand disconnections and sign up topics in schizophrenia because that's where you know that's where i came from in psychiatry um then it's about now that people are starting to take this technology seriously and start to apply it um at scale and in a way that is underwritten by companion studies of the same systems and different models in cell cultures for example so you know there are if you've got one generative model under the hood that explains disconnection of a certain kind due to this synaptopathy which we're now associating with abnormal precision control of exactly the same kind that mark was intimating when he's talking about his broken midbrains because these are the ones that broadcast the signals that control the synaptic efficacy uh you're in exactly the same way as you might imagine that happening in parkinson's disease where you've got abnormal um you're a hypodopenergetic the complement in schizophrenia being a high potentially hyper dopamergic state and many other conditions um so those that that application of dynamic causal modelling at different scales and across scales is now starting to emerge at least in proposal form um no one yet has had the nerve to fund it my hope is in the next year or two there are sufficient number there are a sufficient number of proof of principle that it now becomes uh you know a viable uh way of proceeding and to bring things to closure the two areas that seem to that have really exploited um way of making sense of data to get at synaptopathies are epilepsy research in particular childhood epilepsy and some neurodegenerative conditions so that that's where the action is at the moment yeah i could definitely see that um i want to ask just just as a since this is your kind of more closely you're closer to this is about fmris um just as a small side here many people were very excited about fmris when they came out you know this is very literally colorful exciting in some ways but over time i guess over 30 years now i mean there are some people that would say that it's a kind of you know modern day phrenology do you agree with that what's the utility of fmris if anything and you know how good is it really at translating into i guess better treatments you know because you know some area because we know with diffusion tensor imaging that there's all of the brain is connected all of it is linked up all of it is is talking with each other different parts of the brain cortical subcortical different lobes everything's talking to each other how do we understand i guess the utility of fmri's if at all um and and how how could it translate i guess into to better treatments so you um saying some people are starting to call it a

sort of latter day uh phrenology if they were calling it that at the inception [Laughter] the new cartography the new the neon phenomenon right um which is a fair a fair critique um so i think the the answer to that question is a little bit like this you know how far we've come in our journey if you ask year by year yes the the advances have been incremental at best but if you just look back um with a temper resolution of decades fmri has had quite a profound impact on the clinical cognitive and translational neurosciences so simple things which people of your gentle and tender age will not be aware of simple things that you take for granted were just hypotheses in 1989 1990 for example the whole notion of functional specialization in the brain that underwrote sort of fedorian modalities and and all sorts of hypotheses about um in fact also subtended neophrenology the notion that one part of the brain did this and another part of the brain did that so this is a notion of functional specialization in particular segregation that this particular functionality was segregated anatomically or spatially to this part of the brain uh and that therefore we had at least a good hypothesis about the predictions that would ensue foreign brain damage and we can make some guesses about the importance of connectivity under the functional segregation hypothesis no one knew though no one up until 1990 you know 1989 knew what no one knew that um there was for example um color specialization in this particular part of v4 um that was segregated in the sense that there was no color related responses elsewhere you couldn't do that because you'd have to have electrodes everywhere in the in order to show that it was this part of the brain responding to color and not that part of the brain so it was only in in the early 1990s that people could actually after about 200 years possibly 150 years um confirm the original hypothesis about functional localization specialization and segregation in the brain that was a profound thing as i say from your perspective you you know you just assume that people in the 1980s believe that functional specialization was the principle of brain organization they did believe it but it was just a and there was doubt after fmri there was no doubt it was just part of the foundation of neophenology the second sort of big point to make is that you know if you want to understand how a virus spreads um or how a meme spreads or how um messages spread in a deep neural network or how belief updating operates in a functionally segregated specialized hierarchical you have to have a way of identifying the things that are connected so if you are truly a cartographer when you go to a new country the first thing you do is identify the landmarks before you start worrying about the communication the rivers or indeed if it's a a developed uh country the roots of communication the rail links and all of that good stuff the first thing you do is have to identify what's happening where in terms of these functionally segregated um uh

definitive aspects of any map or before you can talk about distributed pressing well distributed about amongst what well amongst the functionally specialized areas where are they we can now identify those using fmri but of course that's just the first step so within about sort of five to eight years people were turning their attention to the real question which is back to the connectivity it's about it's back to the principles that underwrite the distributed presence saying as you say understanding why every part of the basis could be connected at least with two degrees of freedom to every other part of the brain so that that's where the big questions were and um subsequently addressed in terms of fundamental distinctions between things like functional effective connectivity there was um you know a period when all the excitement was about some measuring correlations and and functional connectivities in the brain um but uh throughout that period of course you things like dynamic cause and modeling were being developed in order to measure the actual uh directed coupling that subtended these correlations that people measuring with functional connectivity was fmri or is fmri useful in that regard and has it had any material effect on um sort of treatment i i i mean i can cite particular examples in terms of epileptic research and identification of sort of eloquent areas the use of imaging as a biomarkers in uh say genetic studies of predisposition to various illnesses you know i could give you a number of some um so quite compelling but very um colloquial examples of where it has worked i would i would sort of concur with you though there hasn't been an awakening of insights it's more a consolidation of what we have known from centuries of careful neurological and neurophysiology an ever-increasing fine-grainedness that allows us now to talk about the particular subreceptors involved in a particular syndrome that underwrite the synaptic or the autopathy that's producing the the disconnection in your in this particular system so the interceptive system involving the anterior cingulate and the anterior insula so it's really people really drilling down on the fine grain in certain quarters of your energy there's a lot of be great stuff as well um but you know the a grade uh mechanistically uh mechanistic um theory-driven applications hypothesis-driven applications of neuroenergy you know really has advanced quite quite remarkably on in the basic neurosciences that you know whether that has yet to translate into um clinical neuroscience you know i think that's an open question but you know the answer i think will only reveal itself when you look back and say what's happening the past 10 years and you probably get a much more positive response than otherwise just one final comment and it certainly is the case that fmri an imaging modality has lost its shine in the sense of its appalling temporal resolution um which means that most earnest imaging neuroscientists are

or cognitive neuroscientists or indeed clinical neuroscientists would want to supplement the the cartographic gift of fmri with highly resolved electromagnetic electrophysiological responses so you know anybody doing any serious imaging neuroscience nowadays would also recourse to electromagnetic signals and standing distributed processing at that level guided by the cartography on offer from fmri yeah no that's that's that's important to to to also uh uh keep in mind as well well my my my last question for you i guess sort of a sort of a big one uh there's two parts to it so the first part is you've talked about many of your um theoretical uh principles you've talked about some of the stuff from the the imaging techniques so we've talked about the bayesian brain which has some utility or has many utilities but many people are are starting to talk more about it mainstream and have been for for many years but in terms of your free energy principle and how you use this with active inference inference excuse me predictive coding theory how do how do you think this can be translated to the common person right how can people use these wonderful uh theoretical structures and frameworks in a way that's applied and has some pragmatic utility and ways that everyone can kind of grasp and say oh yes i got that and i see how this is really important or use it in various uh uh clinical practices in whichever domain in psychiatry neuroscience and psychology et cetera how how do you think um the ability to translate those models and theories into practical applied use is is is key and and and ways to do that larger frame or larger or of this is for neuroscience as a whole you know how do you see i mean neuroscience you know its utilities maybe some of its limitations but how it's it's useful for for people to continue to understand the complexities of neuroscience in a sense we've already sort of covered i think the the two key um practical applications um and perhaps it would probably just to rehearse the point that there's a lot of similarity between the free energy principle and the theory of natural selection you know in and of themselves they do nothing for you you've actually got to want to answer a question in a particular situation or to build something um you know in a given circumstance um and then the other free energy principle and its attendant process there is will certainly provide you the functional forms and the architecture the mechanics to do that quite quickly and efficiently um so the question now is why would you ever want to build those things and how could you practically use them um so in terms of um building sentient artifacts you what you know i just remind um myself and you know recall an early part of the conversation that what we have now is a an off-the-shelf set of belief updating schemes that should in principle um create curious sentient robots and artifacts that will be um to my mind the next step in artificial or machine learning and i say that from a number of perspectives first

of all in terms of you know what's um what what is a use case that would be appreciated by members of the public i i generally think that it's probably going to be the manufacture of little pets as much of sort of you know public facing ai has pursued but the difference will be here these artificial pets will actually have a curiosity and it will be some it will be much easier to um project and anthropomorphize and to apply theory of mind to these kinds of little curious pets and so they'll become increasingly um more like your actual favorite uh pet or your your or your dog simply because they now have um curiosity under the hood so that's how i personally think people are going to make money out of active inferences just equipping these little robots with a bit of curiosity or to use jurgen schmidt he was phrasing artificial curiosity um and interestingly mentioning yogurt of course he you know he thinks that's the essence of creativity and fun it's this is curiosity this epistemic drive this information seeking part of resolving uncertainty or maximizing information again through actively uh soliciting evidence for your generative models of in terms of machine learning and the more i think pressing a less trivial issue of sustainability the whole point of active inference is that it is a bayes optimal in the way that it gets data and assimilates those data and that optimality expresses itself in certain kinds of efficiency which i think are going to be very important in terms of sustainable computation and energy use so i've heard that in 2025 about 20 percent if not 22 the total energy consumption globally will be um by data farms and uh you know e-commerce and exchange so this is a really if you like thermodynamically and climate change relevant observation there are good ways to make imprints and to data mine and there are bad ways now the good ways are those that conform to hamilton's principle at least action that's the free energy principle that you minimize in your process of inference or active inference you maximize the evidence which mathematically simply means minimizing the complexity under accuracy constraints where the complexity is the the cost you pay in terms of belief updating is literally that measure of movement from the prior to the post area that costs energy via the zheninski equality which means if you use the wrong kind of overly parametrized generative model say with a billion parameters in a state-of-the-art learning network you are doing it in the statistically least optimal way and using a massive amount of electricity so another way of stating that is the direction of travel in machine learning in particular deep learning is exactly the wrong direction because it relies upon big data an overly parameterized overly complex charitable so i'm hoping that active inference in the free energy principle will reverse that direction of travel through proof of principle so what we have instead now is not throwing billions of data points at your deep learning machine and millions of dollars to a team of um

engineers who are trying to handcraft and tune the parameters of the of their particular variational auto encoder along short-term memory layers in you know a convolutional network instead you will spend you will let the machine itself decide what is the most informative data it needs to resolve its uncertainty it will run either on an edge computing device but it's only running your laptop and it should finish and converge within minutes to days as opposed to days to months and it will cost you a hundred dollars or even ten dollars as opposed to up to a million dollars so i'm deliberately completely overstating the benefits but that reversal or direction of travel um is mathematically um uh uh obvious it hits you in the face when you start to sort of articulate computation intelligence in terms of the process of inference and just look at the underlying maths in terms of the information geometry and then using things like the jinski equality to understand that in terms of the both the computational cost but also the thermodynamic cost of computation so i think that's going to be in principle i'm sure i'm completely wrong i'm probably talking complete rubbish but in my little fantasy world data mining is a thing of the past you won't need data farms because instead of going for mining big data these curious machines will go in mind just the sparse data they need to resolve their uncertainty for what they are um for the context in which they operate so we'll see a move to edge computing with sparse data but it will be bayes optimal in the sense it's just returning to the basic principles of optimal basic design established in the 1950s by people like david lindley sort of revisited by people like david mackay in terms of active learning in the 1990s and then apparently completely forgotten about uh until the active inference movement or push back against the direction of travel and sort of standard deep learning approaches or even reinforcement learning without the it's it's it's it's incredible to think about all of the avenues that it that it that this has the potential to to expand uh with well i mean i i don't know what to say uh carl you've been so so generous uh i mean more than generous with with your time and i i consider it a great privilege and honor that you have given me over three hours of your time to explain this that's that's quite the gift and i greatly appreciate that is there any final comments that you want to say or anything that you want to summarize or anything like that and and then also i'm very impressed with the old staliner stamina and also there's some you know the patience with which you ask your questions wait for about 20 minutes for for example of an answer that that is quite remarkable so thank you no i the the pleasure is is all mine you have been very very generous and i i am quite sure that listeners will be uh very very thrilled to to hear all of your wonderful work that you've been doing so um can't say enough thanks appreciate it you